Chulwoo Pack

Assistant Professor, Electrical Engineering & Computer Science, South Dakota State University

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Research Interests

Multimodal AI, Information Retrieval, Digital Library, Anomaly Detection, Multimedia Analysis, eXplainable AI.

Current Position

Assistant Professor, McComish Department of Electrical Engineering & Computer Science, South Dakota State University, Brookings, SD — Aug 2023–Present

Education

Ph.D. in Computer Science, University of Nebraska–Lincoln, Lincoln, NE — 2023
Dissertation: Enhancing Document Layout Analysis on Historical Newspapers: Visual Representation, Pseudo-ground-truth, and Downscaling
Advisor: Leen-Kiat Soh

M.S. in Computer Science, South Dakota State University, Brookings, SD — 2017
Thesis: Optimizing Multilayer Perceptron with Dynamic Learning Rate to Classify Breast Microwave Tomography
Advisor: Sung Shin

B.S. in Computer Science, South Dakota State University, Brookings, SD — 2015
Magna cum laude

B.Eng. in Computer Engineering, University of Ulsan, Ulsan, Korea — 2015
Dual degree with South Dakota State University

Experience

Graduate Research Assistant, Center for Brain, Biology, and Behavior, University of Nebraska–Lincoln — Sep 2021–May 2023

Graduate Research Assistant, Cyber-Security Education, University of Nebraska–Lincoln — Jan 2021–Aug 2021

Research Scientist/Engineer Intern, Digital Strategy Division, Library of Congress — Jul 2019–Jan 2020

Graduate Research Assistant, Image Analysis for Archival Discovery (AIDA), University of Nebraska–Lincoln — Dec 2017–Dec 2020

Graduate Research Assistant, Convergent Computing Technology Laboratory, South Dakota State University — Jan 2015–Mar 2016

Student Web Developer, Office of Information & Technology, South Dakota State University — May 2013–May 2017

Grants Awarded

[g6] **Co-PI**, “Center for AI-Driven Water, Energy, and Agricultural Resilience in the Upper Midwest,” SDSU Interdisciplinary Research Clusters, May 2026–May 2027, $30,000.

[g5] **Co-PI**, “SAFE-AG Cluster: AI-Enabled Vital Sign and Exposure Monitoring to Advance Rural Agricultural Worker Health,” SDSU Interdisciplinary Research Clusters, May 2026–May 2027, $15,000.

[g4] **Co-I**, “Collaborative Research: FEC: Establishing Infrastructure for AI-Driven Discovery of Small Molecules to Combat Antibiotic Resistance, Biofilms, and Aflatoxin Contamination,” NSF RII, Aug. 2025–Jul. 2029, $2,190,786.

[g3] **Co-PI**, “Digital Light Processing (DLP)-Based Hyperspectral Imaging System using AI Edge Computing for Smart Agricultural Applications,” SDSU Jackrabbit Jumpstart Grant Program, Jul. 2025–Aug. 2026, $14,358.

[g2] **PI**, “Explainable AI for Safer Construction: Vision-Based Fall Detection and Hazard Mitigation,” SDSU RSCA, Jul. 2025–Aug. 2026, $15,000.

[g1] **Co-PI**, “RAISE Program: Robotics, AI, and Sensor Education in K-14 Agriculture & STEM,” USDA NIFA, Jun. 2024–May 2026, $155,310; supplemental funding awarded in 2026: $150,752.

Publications

Refereed Conference Papers

[c15] Sainath Reddy Gummi, **Chulwoo Pack**, Hankui Zhang, Shyam Solanki, and Young Chang. “Towards Morphology Aware Stomata Keypoint Detection: Benchmarking Foundation Models Under Distribution Shift.” In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9730-9739. 2026.

[c14] Jingong Chen, Dongyoun Kim, **Chulwoo Pack**, Jun Huang, and Kwanghee Won. “TIU-ReID: Target Identity Unlearning for Person Re-Identification.” In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8506-8515. 2026.

[c13] Mukhtiar Ali, and **Chulwoo Pack**. “Multi-Scale Encoding, Gated Fusion, and Temporal Context for Lightweight Wearable Stress Detection.” In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7454-7462. 2026.

[c12] Muhammad Roman, Mazhar Sher, Chamika Kuruppuarachchi, Arshid Ali, **Chulwoo Pack**, Azlan Zahid, and Ali Mirzakhani Nafchi. “Deep learning-based classification of nitrate and nitrite concentrations from water samples using colorimetric test strip images.” Environmental Monitoring and Assessment 198, no. 5, 447, 2026.

[c11] Omeshamisu Anigala, Joy Nnenna Okolo, Junggab Son, and **Chulwoo Pack**, “MM-SCORE: A Two-Factor Framework for Auditing Multimodal Dataset Quality,” Proceedings of the International Conference on Research in Adaptive and Convergent Systems, pp. 1–8, 2025.

[c10] Harsh Dubey and **Chulwoo Pack**, “Leveraging Textual Memory and Key Frame Reasoning for Full Video Understanding Using Off-the-Shelf LLMs and VLMs (Student Abstract),” Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 28, pp. 29351–29352, 2025.

[c9] Hasan Mirzakhaninafchi, **Chulwoo Pack**, Dongyoun Kim, Young Chang, and Kwanghee Won, “Comparative Analysis of Deep Learning-Based Anomaly Detection Models for GPS Spoofing Detection,” Proceedings of the International Conference on Research in Adaptive and Convergent Systems, pp. 113–120, 2024.

- [c8] Youngsun Jang, Dongyoun Kim, **Chulwoo Pack**, and Kwanghee Won, "A Novel Dataset for Flood Detection Robust to Seasonal Changes in Satellite Imagery," Proceedings of the International Conference on Research in Adaptive and Convergent Systems, pp. 177–184, 2024.
- [c7] Sainath Reddy Gummi, **Chulwoo Pack**, Pappu Yadav, Young Chang, Mazhar Sher, James Kemeshi, and Mohammad Ashik Alahe, "A Deep Learning Approach for Precision Phenotyping of Corn Stomata," in Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping IX, vol. 13053, pp. 66–73, 2024.
- [c6] **Chulwoo Pack** and Dongyoun Kim, "Weakly-supervised Semantic Segmentation on Historical Document Images," Proceedings of the Conference on Research in Adaptive and Convergent Systems, pp. 1–6, 2023.
- [c5] Rui Zhao, Harvey Siy, **Chulwoo Pack**, Leen-Kiat Soh, and Myoungkyu Song, "An Intelligent Tutoring System for API Misuse Correction by Instant Quality Feedback," Proceedings of the 45th Annual Computers, Software, and Applications Conference, pp. 123–128, 2022.
- [c4] **Chulwoo Pack**, Sung Shin, Hyung-Do Choi, Soon-Ik Jeon, and John Kim, "Optimized Multilayer Perceptron Using Dynamic Learning Rate-based Microwave Tomography Breast Cancer Screening," Proceedings of the 31st Annual ACM Symposium on Applied Computing, pp. 2171–2175, 2016.
- [c3] **Chulwoo Pack**, Sung Shin, Seong-Ho Son, and Soon-Ik Jeon, "Computer Aided Breast Cancer Diagnosis System with Fuzzy Multiple-parameter Support Vector Machine," Proceedings of the Conference on Research in Adaptive and Convergent Systems, pp. 172–176, 2015.
- [c2] Sallies Gc, **Chulwoo Pack**, Sung Shin, and Hyung-Do Choi, "Breast Cancer Classification of Mammography Masses Using Improved Shape Features," Proceedings of the Conference on Research in Adaptive and Convergent Systems, pp. 188–194, 2015.
- [c1] Samaneh Aminikhanghahi, Sung Shin, Wei Wang, Soon-Ik Jeon, Seong-Ho Son, and **Chulwoo Pack**, "Study of Wireless Mammogram Image Transmission Impacts on Robust Cyber-aided Diagnosis Systems," Proceedings of the 30th Annual ACM Symposium on Applied Computing, pp. 2252–2256, 2015.

Refereed Journal Articles

- [j7] Omeshamisu Anigala, Kwanghee Won, and **Chulwoo Pack**, "PEARL: Perceptual and Analytical Representation Learning for Video Anomaly Detection," ACM SIGAPP Applied Computing Review, vol. 25, no. 1, pp. 5–15, 2025.
- [j6] Leen-Kiat Soh, Elizabeth Lorang, **Chulwoo Pack**, and Yi Liu, "Applying Image Analysis and Machine Learning to Historical Newspaper Collections," The American Historical Review, vol. 128, no. 3, pp. 1382–1389, 2023.
- [j5] **Chulwoo Pack**, Leen-Kiat Soh, and Elizabeth Lorang, "Perceptual Cue-Guided Adaptive Image Downscaling for Enhanced Semantic Segmentation on Large Document Images," International Journal on Document Analysis and Recognition, pp. 1–17, 2023.
- [j4] **Chulwoo Pack**, Leen-Kiat Soh, and Elizabeth Lorang, "Visual Domain Knowledge-based Multimodal Zoning for Textual Region Localization in Noisy Historical Document Images," Journal of Electronic Imaging, vol. 30, no. 6, 2021.
- [j3] **Chulwoo Pack**, Yi Liu, Leen-Kiat Soh, and Elizabeth Lorang, "Augmentation-based Pseudo-Groundtruth Generation for Deep Learning in Historical Document Segmentation for Greater Levels of Archival Description and Access," Journal of Computing and Cultural Heritage, 2021.
- [j2] **Chulwoo Pack**, Seong-Ho Son, and Sung Shin, "Computer Aided Diagnosis with Boosted Learning for Anomaly Detection in Microwave Tomography," ACM SIGAPP Applied Computing Review,

vol. 17, no. 3, pp. 39–47, 2017.

[j1] **Chulwoo Pack**, Samaneh Aminikhanghahi, Sung Shin, Soon-Ik Jeon, and Seong-Ho Son, "An Optimized Fuzzy Support Vector Machine Classifier Using Breast Mammogram Tomography: Trade-off Between Specificity and Sensitivity," International Information Institute, vol. 18, no. 9, pp. 3979–3988, 2015.

Other Scholarly Publications and Reports

[o2] Elizabeth Lorang, Leen-Kiat Soh, Yi Liu, and **Chulwoo Pack**, "Digital Libraries, Intelligent Data Analytics, and Augmented Description: A Demonstrated Project," submitted to the Library of Congress, 2020.

[o1] Elizabeth Lorang, Leen-Kiat Soh, **Chulwoo Pack**, and Yi Liu, "Application of the Image Analysis for Archival Discovery Team's First-Generation Methods and Software to the Burney Collection of British Newspapers," CDRH Grant Reports, no. 7, 2019.

Presentations

Invited Talks

[t3] **Invited Speaker**, "AI in Digital Library," Briggs Library Professional Development, South Dakota State University — Mar 2026

[t2] **Invited Speaker & Panelist**, "Computer Vision, Deep Learning, and Reasoning in Document Understanding," 10th Annual Nebraska Forum on Digital Humanities, University of Nebraska–Lincoln — Apr 2024

[t1] **Guest Lecturer**, "Computer Vision Applications Around Us," University of Nebraska–Lincoln — Nov 2023

Research Presentations

Research Presenter, Korean Computer Scientists and Engineers Association in America (KOCSEA), Las Vegas, NV — Nov 2025

Research Presenter, Digital Strategy Division, Library of Congress, Washington, DC — Jan 2020

Research Presenter, National Digital Newspaper Program Awardee Conference, Washington, DC — Sep 2018

Poster Presenter, Sanford Health–SDSU Biomedical Research Symposium, Sioux Falls, SD — Nov 2015

Professional Service

Committee

Workshop Chair, ACM Research in Adaptive and Convergent Systems — 2025–2026

Program Committee, AAAI — 2026

Editorial Review Board, Construction Safety Conference: Safety Knowledge Foundations — 2025

Undergraduate Curriculum Committee Chair, South Dakota State University — 2025–2026

Judge, SDSU Jackrabbit BEST Robotics — 2015

Assistant staff, SDSU Programming Design Competition — 2014–2016

Paper Reviewer

Conferences: ICML, AAAI, WACV, ICDAR
Journals: *IEEE Sensors*, *IEEE Transactions on Geoscience and Remote Sensing*, *International Journal of Multimedia Information Retrieval*, *Journal of Supercomputing*

Teaching Experience

Instructor, Introduction to Deep Learning (CSC 492/592), South Dakota State University — SP 2026
Instructor, Machine Learning Systems Design (CSC 792), South Dakota State University — FA 2025
Instructor, Foundations of Software Engineering (SE 305), South Dakota State University — FA 2025
Instructor, Introduction to Deep Learning (CSC 492/592), South Dakota State University — SP 2025
Instructor, Software Engineering Management (CSC 770), South Dakota State University — FA 2024
Instructor, Computer Vision and Pattern Recognition (CSC 492/592), South Dakota State University — SP 2024
Instructor, Software Engineering Management (CSC 770), South Dakota State University — SP 2024
Teaching Assistant, Automata, Computation and Formal Languages (CSCE 428/528), University of Nebraska–Lincoln — FA 2017

Honors and Awards

Best Paper Award, SPIE Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping IX — 2024
Student Travel Award, ACM Symposium On Applied Computing — 2016
Overseas Study Scholarship, UOU Foundation — 2011
Great Enrollment Scholarship, Hyundai Heavy Industries — 2011–2014

Student Mentoring

Ph.D. Students

Harsh Dubey — 2026–Present
Mukhtiar Ali — 2024–Present
Omesamisu Anigala *(Current: Software Engineer at OIT, SDSU)* — 2023–2025

M.S. Students

Preethi Amasa — 2025–Present
Mehul Deep — 2025–Present
Somtochukwu Orizu — 2025–Present
Hossein Erfanshekooh — 2024–Present
Sugam Mishura — 2024–2025
Harsh Dubey — 2024–2025

Farzaneh Karimpour 2023–2024

Undergraduate Students

Kable Richard Morrill 2026–Present

John Akujobi *(Future Innovators of America Awardee; Undergraduate Summer Research Awardee)* 2025–Present